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Program-3: Code a Dockerized Python Flask or Node.js Application

Structure of the program:

Dockerfile
app.py
Requirements.txt

Step 1:
Create a directory and name it program-3 and write all the files required as shown in structure of the program

```
_1RV24MC019_anusha_s@anu-Vivobook-Go-E1504FA-E1504FAB:~$ mkdir progra-m
_1RV24MC019_anusha_s@anu-Vivobook-Go-E1504FA-E1504FAB:~$ cd progra-m/
_1RV24MC019_anusha_s@anu-Vivobook-Go-E1504FA-E1504FAB:~/progra-m$ nano Dockerfile
_1RV24MC019_anusha_s@anu-Vivobook-Go-E1504FA-E1504FAB:~/progra-m$ nano app.py
_1RV24MC019_anusha_s@anu-Vivobook-Go-E1504FA-E1504FAB:~/progra-m$ nano requirements.txt
```

Step 2:
Create a docker file -which contains the instructions to create a docker image

```
#Uses the official python image
FROM python:latest

#Set the working directory
WORKDIR /app

#Copy the requirements file and install the dependenices
COPY requirements.txt .
RUN pip install --no-cache-dir -r requirements.txt

#Copy the application code
COPY . .

#Expose the port
EXPOSE 5000

#Run the application
CMD ["python","app.py"]
```

Step 3:
Create a file app.py and add all the following code

```
from flask import Flask

app=Flask(__name__)

@app.route("/")
def home():
    return "Welcome to docker"
if __name__=="__main__":
   app.run(host="0.0.0.0",port=5000)
```

Step 4:
Create a requirements.txt and add the following

```
Flask==2.3.3
```

Step 5:

Build the image called program-3

```
_1RV24MC019_anusha_s@anu-Vivobook-Go-E1504FA-E1504FAB:~/progra-m$ sudo docker build -t program-3 .
[sudo] password for _1RV24MC019_anusha_s:
[+] Building 0.2s (10/10) FINISHED
 => [internal] load build definition from Dockerfile
 => => transferring dockerfile: 378B
 => [internal] load metadata for docker.io/library/python:latest
 => [internal] load .dockerignore
 => => transferring context: 2B
 => [1/5] FROM docker.io/library/python:latest
 => [internal] load build context
 => => transferring context: 640B
 => CACHED [2/5] WORKDIR /app
 => CACHED [3/5] COPY requirements.txt .
 => CACHED [4/5] RUN pip install --no-cache-dir -r requirements.txt
 => [5/5] COPY . .
 => exporting to image
 => => exporting layers
 => => writing image sha256:3b1ee176e6566e259aedd8426b3034703e48367d6ce76d664c34168e65b8effc
 => => naming to docker.io/library/program-3
```

Step 6:

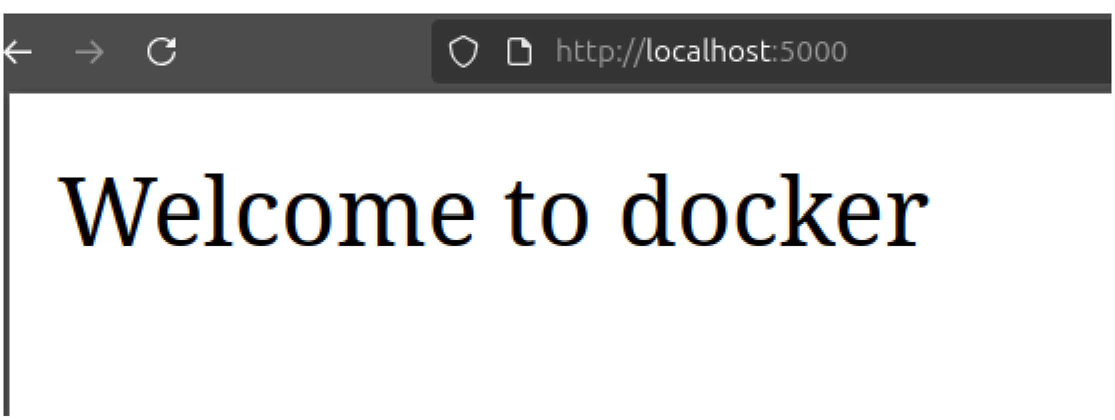
Build the container with the image program-3

```
_1RV24MC019_anusha_s@anu-Vivobook-Go-E1504FA-E1504FAB:~/progra-m$ sudo docker run -p 5000:5000 --name container-3 program-3
 * Serving Flask app 'app'
 * Debug mode: off
WARNING: This is a development server. Do not use it in a production deployment. Use a production WSGI server instead.
 * Running on all addresses (0.0.0.0)
 * Running on http://127.0.0.1:5000
 * Running on http://172.17.0.2:5000
Press CTRL+C to quit
172.17.0.1 - - [07/Nov/2025 03:48:01] "GET / HTTP/1.1" 200 -
```

Step 7:

To check the container go to browser and navigate to http://localhost:5000/

You will see “Welcome to Docker” message



Step 8:

View all the container and their respective images and then stop,remove the containers

```
^C_1RV24MC019_anusha_s@anu-Vivobook-Go-E1504FA-E1504FAB:~/progra-m$ sudo docker container ls -a
CONTAINER ID   IMAGE       COMMAND           CREATED          STATUS                      PORTS     NAMES
f8cf6d70d9a1   program-3   "python app.py"   51 seconds ago   Exited (0) 11 seconds ago             container-3
_1RV24MC019_anusha_s@anu-Vivobook-Go-E1504FA-E1504FAB:~/progra-m$ nano Dockerfile
_1RV24MC019_anusha_s@anu-Vivobook-Go-E1504FA-E1504FAB:~/progra-m$ nano app.py
_1RV24MC019_anusha_s@anu-Vivobook-Go-E1504FA-E1504FAB:~/progra-m$ nano requirements.txt
_1RV24MC019_anusha_s@anu-Vivobook-Go-E1504FA-E1504FAB:~/progra-m$ sudo docker container ls -a
[sudo] password for _1RV24MC019_anusha_s:
CONTAINER ID   IMAGE       COMMAND           CREATED          STATUS                      PORTS     NAMES
f8cf6d70d9a1   program-3   "python app.py"   21 minutes ago   Exited (0) 21 minutes ago             container-3
_1RV24MC019_anusha_s@anu-Vivobook-Go-E1504FA-E1504FAB:~/progra-m$ sudo docker container stop f8c
f8c
_1RV24MC019_anusha_s@anu-Vivobook-Go-E1504FA-E1504FAB:~/progra-m$ sudo docker container rm f8c
f8c
_1RV24MC019_anusha_s@anu-Vivobook-Go-E1504FA-E1504FAB:~/progra-m$ sudo docker images
REPOSITORY           TAG         IMAGE ID       CREATED          SIZE
program-3            latest      3b1ee176e656   23 minutes ago   1.13GB
python               latest      0134b02fcae7   2 days ago       1.12GB
mynginx              latest      1d7074189d7e   3 days ago       150MB
node                 20-alpine   2b56f2779663   3 weeks ago      134MB
mongo                7           ad5e46b29e1a   4 weeks ago      834MB
ubuntu               latest      97bed23a3497   5 weeks ago      78.1MB
registry             2           26b2eb03618e   2 years ago      25.4MB
mysql/mysql-server   5.6         8c587c89edee   4 years ago      238MB
```

Step 9:

Remove the image after the use

```
_1RV24MC019_anusha_s@anu-Vivobook-Go-E1504FA-E1504FAB:~/progra-m$ sudo docker image rm 3b1
Untagged: program-3:latest
Deleted: sha256:3b1ee176e6566e259aedd8426b3034703e48367d6ce76d664c34168e65b8effc
_1RV24MC019_anusha_s@anu-Vivobook-Go-E1504FA-E1504FAB:~/progra-m$
```